

Java includes a rich set of built-in classes provided by its standard library, commonly referred to as the **Java API**. These classes are organized into **packages** such as `java.lang`, `java.util`, `java.io`, and more. Below is an overview of some key packages and their commonly used classes:

1. `java.lang` Package

The `java.lang` package is automatically imported in every Java program. It contains fundamental classes.

- `Object` : The root of the class hierarchy; every class in Java inherits from `Object`.
 - `String` : Represents sequences of characters.
 - `StringBuilder` / `StringBuffer` : For mutable strings.
 - `Math` : Provides mathematical operations and constants.
 - `System` : Contains system utilities like input/output and garbage collection.
 - `Thread` : For multithreading.
 - `Runtime` : Deals with the Java runtime environment.
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2. `java.util` Package

Contains utility classes for data structures, collections, and date/time manipulation.

- **Collection Framework:**
 - `ArrayList`, `LinkedList`, `HashSet`, `TreeSet`, `HashMap`, `TreeMap`, `PriorityQueue`
- **Date/Time API:**
 - `Date`, `Calendar`, `LocalDate`, `LocalTime`, `LocalDateTime`, `ZoneId`
- **Other Utilities:**

- `Scanner` : For input parsing.
 - `Random` : For generating random numbers.
 - `Arrays` : For array-related operations.
 - `Collections` : For collection-related utilities.
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3. `java.io` Package

Contains classes for input/output operations.

- **File Operations:**
 - `File` : Represents file and directory paths.
 - `FileReader` / `FileWriter` : For reading/writing text files.
 - **Stream API:**
 - `InputStream` , `OutputStream` : For byte-stream operations.
 - `BufferedReader` , `BufferedWriter` : For efficient reading/writing.
 - `ObjectInputStream` , `ObjectOutputStream` : For object serialization.
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4. `java.nio` Package

Used for non-blocking I/O and advanced file operations.

- `Paths` and `Files` : Modern ways to work with file systems.
 - `ByteBuffer` , `CharBuffer` : For buffer management.
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5. `java.net` Package

Contains classes for networking.

- `URL` : Represents a Uniform Resource Locator.

- `URLConnection` : For HTTP connections.
 - `Socket` , `ServerSocket` : For socket programming.
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6. `java.sql` Package

Classes for database interaction.

- `Connection` : Represents a connection to a database.
 - `Statement` , `PreparedStatement` : Execute SQL queries.
 - `ResultSet` : Retrieve and manipulate query results.
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7. `java.time` Package (Java 8 and later)

A modern date/time API.

- `LocalDate` , `LocalTime` , `LocalDateTime` : Immutable date-time objects.
 - `Instant` , `Duration` , `Period` : Represent time intervals.
 - `DateTimeFormatter` : For formatting dates and times.
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8. `java.util.concurrent` Package

For concurrent programming and multi-threading.

- `Executor` , `ExecutorService` : Manage threads.
 - `Future` : Handle asynchronous results.
 - `ConcurrentHashMap` , `CopyOnWriteArrayList` : Thread-safe collections.
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9. `javax` Packages

Various packages beginning with `javax` include:

- `javax.swing` : For creating GUI applications.
 - `javax.servlet` : For web applications.
 - `javax.xml` : For XML parsing and processing.
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Example Usage

java

```
import java.util.ArrayList;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = scanner.nextInt();

        ArrayList<Integer> list = new ArrayList<>();
        list.add(num);
        System.out.println("List: " + list);
    }
}
```

Java's built-in classes help you handle tasks ranging from basic string manipulation to advanced concurrency and networking. You can refer to the [Java Documentation](#) for detailed information on these classes.